Title Page

Article type: Original research (hypothesis-driven research)

Title: Segmentation, type classification and measurement of brain tumors on single 2D MRI slices: a retrospective development and leakage-controlled internal validation study

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Statements and Declarations

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Author contributions: Debobrato Biswas conceived and designed the study, implemented the software, performed the experiments and analysis, and wrote and revised the manuscript. The author read and approved the final manuscript.

Ethics approval: Not applicable. This retrospective study used only publicly available, de-identified datasets released by their creators for research (figshare brain tumor dataset; Kaggle Brain Tumor MRI Dataset); no new data were collected from humans.

Consent to participate: Not applicable (public, de-identified data).

Consent to publish: Not applicable. The images shown contain no identifying information.

Data availability: The datasets analyzed in this study are available in the figshare repository (https://doi.org/10.6084/m9.figshare.1512427) and the Kaggle repository (https://doi.org/10.34740/KAGGLE/DSV/2645886). The patient-level split and the per-image results are provided as Online Resources 2–4.

Code availability: The source code, with a user guide, is available at https://github.com/Debobrato06/BrainTumorDetection (archived version **[TBD: Zenodo DOI of the release]**; license **[TBD: e.g. MIT]**). The trained model weights are available from the corresponding author on request.